

TUSHAR BHARDWAJ

Senior Structural / Bridge Engineer • Site Technical Lead & Structural Checking Engineer • Metro, Rail & Highway Infrastructure

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SCAN FOR PORTFOLIO

13+

YEARS IN PRACTICE

7

COUNTRIES DELIVERED IN

405m

LONGEST MAIN SPAN LED

6

DESIGN CODE SYSTEMS

01 PROFESSIONAL SUMMARY

Senior structural and bridge engineer with 13+ years designing, independently checking, and leading on-site technical delivery of complex bridge and metro infrastructure across India, Saudi Arabia, the UK, Australia, Jordan and the UAE. Currently site technical lead on a 405 m main-span cable-stayed bridge in Riyadh — the day-to-day interface between PMC, Designer and subcontractors on lifting, temporary post-tensioning, geometric control, and NCR/FCR resolution — after progressing from Independent Checking Engineer reviewing its superstructure design. Specialises in balanced-cantilever, extradosed, cable-stayed and segmental box-girder bridges. Deeply versed in Indian, AASHTO/ACI, British, Euro and Saudi design codes, with advanced command of Midas Civil, STAAD.Pro, SOFiSTiK and Autodesk Structural Bridge Design.

02 CORE EXPERTISE

- Bridge & Viaduct Structural Design
- Balanced Cantilever, Extradosed & Cable-Stayed Bridges
- Metro Station Substructure & Superstructure Design
- Site Technical Leadership — PMC / Designer / Subcontractor Interface
- Team Leadership, Training & Mentoring
- Independent Checking (ICE) / Proof-Checking
- Segmental & Precast Box Girder Superstructures
- Multi-Disciplinary Coordination (Architecture / MEP / Rail)
- NCR / FCR Resolution & Geometric Control Coordination
- Client & Stakeholder Management

03 TECHNICAL SKILLSET

Software: Midas Civil (Advanced) · STAAD.Pro (Advanced) · SOFiSTiK · Autodesk Structural Bridge Design (Advanced) · OASYS ADSEC (Advanced) · AutoCAD · BIM360 · Excel (Advanced)

Design Codes: IS / IRC / IRS Bridge Code · AASHTO / ACI · British Standards (BS/BD) · Eurocodes · Saudi Building Code · Australian Standards · CIRIA

04 PROFESSIONAL EXPERIENCE

Senior Bridge Engineer

Aug 2025 – Present

ARX Smart Minds | Abu Dhabi, UAE

- Site Technical Lead for the Wadi Laban Cable-Stayed Bridge (179+405+179 m spans, 85+85 m high pylons), Riyadh — the day-to-day technical interface between PMC, Designer and subcontractors
- Resolve issues on lifting & transportation sequencing, temporary post-tensioning, reinforcement/PT clashes, geometrical change and geometric control across the live construction program
- Check and approve shop drawings; raise Field Change Requests (FCRs) and propose engineering resolutions for Non-Conformance Reports (NCRs); close out stakeholder review comments
- Progressed from Independent Checking Engineer (ICE) reviewing superstructure design into this site-leadership role; direct client interactions and design submissions across UAE, Jordan, KSA and Oman rail projects
- Led Structural Proof Checking works for the Footbridge Project in Jubail Golf Estate for Etihad Rail, on behalf of ARX as PMEC

Senior Structural Engineer

Mar 2022 – Aug 2025

Ayesa | Noida, India

- Led design and structural proof-checking for Mumbai Metro Line 2B and Nagpur Metro, including a 70+125.5+75 m balanced cantilever bridge and two extradosed bridges (80 m and 53+104+43 m spans)
- Directed detailed design of Nagpur Metro segmental box girders (spans to 34.8 m); managed design teams for National Water Company projects, Saudi Arabia

Structural Engineer

Nov 2021 – Feb 2022

Vashist Infra Engineering Pvt. Ltd. | New Delhi, India

- Designed substructure and superstructure for the Patna Metro Project, including PSC I-girders, portal frames, piers and pier caps
- Trained and guided junior engineers to meet quality standards and project timelines

Freelance Structural Engineer

Jul 2021 – Oct 2021

Self-Employed

- Developed independent design tools for foundation design and carried out site quality inspections for complex bridge structures

Senior Structural Engineer

Jun 2020 – Jul 2021

Ayesa | New Delhi, India

- Led proof-checking of the Delhi–Meerut RRTS (NCRTC) project: substructure, single-cell box girder superstructure, and balanced cantilever bridge and stations
- Conducted regular site inspections; directed design of tender-variation components to independently verify contractor claims

Structural Engineer

Mar 2019 – May 2020

Ayesa | Riyadh, Saudi Arabia

- Managed structural, architectural and MEP-coordinated design submissions for Riyadh Metro Line 6, proof-checking designs ahead of client and RAMPED (independent checker) review
- Delivered detailed design for structures within a KSA science theme park to AASHTO/ACI standards

Structural Engineer

Oct 2017 – Feb 2019

Jacobs Engineering (Consulting Engineering Services) | Gurugram, India

- Assessed existing UK road and rail bridges, including the 19-span Taf Fechan Viaduct, for revised loading – modelling in Midas Civil and STAAD to BS/BD standards
- Delivered detailed design of Super-T girder highway bridges for the West Gate Tunnel Project, Melbourne, optimised in Autodesk Structural Bridge Design

Assistant Structural Engineer

Nov 2015 – Oct 2017

Ayesa | Noida, India

- Delivered detailed design for the Kolkata Metro viaduct and Dakshineswar / Baranagar stations – piles, pile caps, piers, PSC/RCC/composite girders and steel roof portals
- Designed substructures, including liquefaction analysis, for the Noida–Greater Noida Metro; proof-checked substructure and superstructure packages for Mumbai Metro Line 7

Design & Site Planning Engineer

Jul 2013 – Nov 2015

Larsen & Toubro | New Delhi, India

- Coordinated site-level structural design for DMRC viaducts and elevated stations, checking a 75+120+70 m balanced cantilever bridge and steel-truss erection sequences
- Prepared lifting and launching schemes for PSC I-girders and steel trusses over live railway lines

05 SIGNATURE PROJECTS

PROJECT	STRUCTURE	PRECISE ROLE
Wadi Laban Cable-Stayed Bridge, Riyadh	Cable-stayed, 179+405+179 m spans, 85+85 m high pylons	Site Technical Lead (ex-ICE)
Jordan Rail Project (Phosphate / Potash / Maan Link)	Culverts & minor bridges (Eurocode)	Detailed Design Engineer and Delivery Lead for Structures
Mumbai Metro Line 2B	Balanced cantilever (70+125.5+75 m), 2 extradosed bridges, segmental box girders	Senior Structural Engineer for Proof checking the design
Delhi–Meerut RRTS (NCRTC)	Balanced cantilever bridge, single-cell box girder viaduct, stations	Senior Structural Engineer for Proof checking the design
Riyadh Metro Line 6	Metro stations & viaduct, multi-disciplinary coordination	Design Submission Manager
West Gate Tunnel Project, Melbourne	Super-T girder highway bridges	Detailed Design Engineer
Kolkata Metro	Viaduct & 2 elevated stations, steel roof portals	Assistant Detailed Design Engineer
Noida Metro	Viaduct & 7 elevated stations	Assistant Detailed Design Engineer

06 CERTIFICATIONS, MEMBERSHIPS & EDUCATION

- Pursuing Chartered Professional Engineer (CPEng), Engineers Australia
- Member, Institution of Engineers Australia – ID 9635010 | Member, Indian Association of Structural Engineers – ID M364
- Member, Tunnelling Association of India | Member, Institution of Engineers (India)
- Introduction to Generative AI; Data Science Foundation Course

B.Tech., Civil Engineering – National Institute of Technology, Patna | 2009–2013 | 82.62%